

Description

[0001] The present invention refers to an electrically driven hand-guided power tool, in particular a polisher or a sander. The power tool comprises a tool housing with a recess for receiving at least a part of a battery pack, the design of the recess corresponding to the part of the battery pack inserted therein, in order to allow insertion of the part of the battery pack into the recess and electric connection of the battery pack with the power tool and operation of the power tool by means of electric energy originating from the battery pack.

[0002] Power tools of the above-mentioned kind are well known in the prior art. For example, EP 2 712 713 A1 describes a power tool with a battery pack which is inserted into a recess of the tool housing for operation of the power tool by means of electric energy originating from the battery pack. Battery operated power tools are particularly useful when a high degree of flexibility and moving around of the user with the power tool is required, for example when polishing or sanding small surfaces with a complex three-dimensional extension. Movement of the user and the power tool is not obstructed by an electric cable connecting the power tool to an electronic mains power supply. However, due to a limited capacity of battery packs, battery operated power tools run up against their limits where high power is required over a long period of time, for example when polishing or sanding large surfaces.

[0003] Therefore, it is known from EP 2 712 713 A1 to replace the battery pack with a power supply unit which may be inserted into the recess at least in part instead of the battery pack. The power supply unit is attached to an electronic mains power supply and comprises an electric transformer for transforming the electric energy from the mains power supply into electric energy corresponding to the electric energy for operating the power tool. Hence, when replacing the battery pack with the power supply unit the power tool is operated for an almost infinite amount of time by means of electric energy originating from the mains power supply and transformed by the power supply unit.

[0004] The power supply unit known from EP 2 712 713 A1 is also called "mainsconnector". The term "mainsconnector" refers to a special switching power supply unit for cordless power tools. The power supply unit is in a battery-housing so it can be attached to the power tool just like a conventional battery pack. Hence, a "mainsconnector" allows cordless power tools to be operated with electric energy originating from a mains power supply.

[0005] However, power tools with a power supply unit inserted into the recess of the tool housing have a number of drawbacks. First of all, if a main power supply is developed and sold for insertion into the recess of a tool housing instead of the battery pack, the entire power tool with the power supply unit inserted therein has to be approved (testing and certification) in order to obtain the

approvals (e.g. the VDE-approval) necessary for selling an electric power tool in the European Union. Even though the power tool with the battery pack inserted therein may have already been approved, the replacement of the battery pack by the power supply unit requires a re-approval of the entire power tool together with the power supply unit inserted therein.

[0006] Further, during operation power supply units comprising electric transformers for transforming the electric energy from the mains power supply into electric energy corresponding to the electric energy supplied by the battery pack, a large amount of heat is generated. This is due to the rather low efficiency of known transformers. If the power supply unit is at least partly inserted into the recess of the tool housing heat issues may arise because dissipating the generated heat through the tool housing is difficult and only possible to a limited extent. As a consequence, the power tool may often be automatically turned off as a safety measure in order to avoid overheating of the electric and electronic components of the power tool located inside the tool housing.

[0007] Finally, power supply units comprising electric transformers for transforming the electric energy from the mains power supply into electric energy corresponding to the electric energy supplied by the battery pack are often rather heavy. This results in a shifting of weight of the power tool towards where the recess is provided, usually towards the rear end of the power tool, if the battery pack is replaced by the power supply unit. The power tool may no longer have an even distribution of weight, which may further obstruct ease of use of the power tool.

[0008] Therefore, it is an object of the present invention to provide for an electric power tool which may be operated with a battery pack or with electric energy originating from a mains power supply and which overcomes the above drawbacks. In particular, the power tool should not have to be re-approved if operated with the mains power supply instead of the battery pack, the overheating problem during operation of the power tool with the mains power supply should be overcome and the power tool should have an even distribution of weight no matter if it is operated by the battery pack or with the mains power supply.

[0009] These objects are solved by the electrically driven power tool comprising the features of claim 1. In particular, starting from the electric power tool of the above-mentioned kind, it is suggested that the power tool comprises an adapter element having an adapter housing at least part of which corresponding to the form and design of the recess and adapted to be inserted into the recess after removal of the battery pack, in order to allow electric connection of the adapter element with the power tool, the adapter element being connected to an external power supply unit by means of a first electric cable and the power supply unit being connectable to an electronic mains power supply by means of a second electric cable, the power supply unit comprising an electric transformer for transforming the electric energy from the mains power